

Question number	Answer	Additional guidance	Mark
8(a)	An answer that includes the following point: • (sarcopenia is a progressive) decrease in muscle {mass / function} that is associated with aging (1)		(1)

Question number	Answer	Additional guidance	Mark
8(b)	An explanation that includes the following points: • due to a {specific stimulus / transcription factor / epigenetic modification} (1) • activating (specific / muscle cell / some) genes which are transcribed (1) • translation resulting in the synthesis of named proteins required by muscle cells (1)	accept chemical accept differential gene expression accept mRNA produced from active genes ignore some genes inactivated e.g. actin, myosin, troponin, tropomyosin	(3)

Question number	Answer	Additional guidance	Mark
8(c)	A description that includes the following points: • lower testosterone concentration in the blood (1) • less testosterone {enters (muscle) cell / crosses membrane / present in (muscle) cell nucleus} (1) • forming fewer {hormone-receptor complexes / transcription factors} (1) • (therefore) reduced {transcription / translation} (of muscle protein) (1)	to get full marks they have to mention reduced / less / fewer at least ONCE or it is max 3 marks accept less testosterone in blood accept less testosterone diffuses into (muscle) cell accept fewer muscle protein genes activated accept less testosterone binds to receptor accept less mRNA produced ignore protein synthesis unqualified	(4)

Question number	Answer	Additional guidance	Mark
8(d)	An explanation that includes the following points: • change in condition is detected by a receptor (1) • stimulating an effector that counteracts the change (1)	e.g. {high / low} blood glucose levels detected by (chemo)receptor e.g. pancreas stimulated to release insulin which results in {lower blood glucose / glucose being converted to glycogen} insulin released which results in {lower blood glucose / glucose being converted to glycogen} pancreas releases glucagon which results in {increase in blood glucose} / glycogen being converted to glucose} glucagon released} which results in {increase in blood glucose} / glycogen being converted to glucose}	(2)

Question number	Answer	Additional guidance	Mark
8(e)	An answer that includes two of the following points: - insulin binds to receptors on (muscle) cell (surface) membrane (1) - insulin increases the {number / production / activity} of {glucose transport molecules / glucose channels / GLUT4 (molecules)} (in the cell membrane) (1) - glucose is taken up by (facilitated) diffusion (1)	accept (insulin) stimulates {cAMP / formation of secondary messenger}	(2)

Question number	Answer	Additional guidance	Mark
8(f)	A description that includes the following points: {uses X rays (and computer) to give {cross sectional / 2D / 3D} image (1) {measure / compare} size (on images taken) before and after aerobic exercise (1)		(2)

Question number	Answer	Additional guidance	Mark
8(g)	An explanation that includes the following points: - mitochondria perform {aerobic respiration / produce ATP} (1) {more / more active} mitochondria supply the increased ATP required (for aerobic exercise / increased aerobic respiration) (1) - ATP is needed {to break the cross-bridge / active transport of calcium ions / for muscle contraction} (1)	accept aerobic respiration produces ATP accept more mitochondria supply more ATP more aerobic respiration provides more ATP accept {ATP / energy} needed for protein synthesis	(3)

Question number	Answer	Additional guidance	Mark
8(h)	An explanation that includes three of the following points: (whey / egg protein) contain non-essential amino acids (1) - non-essential amino acids not necessary {for (stimulation of) muscle protein anabolism / for (stimulation of) protein synthesis} (1) (excess) non-essential amino acids can be converted into fats (1) (therefore they) may not {increase / prevent loss of} muscle mass (1)	accept increase of fats increases the risk of cardiovascular disease / diabetes / obesity	(3)

